

Logistic Regression Nuances - Data Analysis and Visualization

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Spring 2025

Classification Cost Function

- Classification output is a categorical variable
- ResSS cannot be used as it is expected
- Instead classification error or fraction of the training observation with error ($\hat{p}_k$) can be used.

$$E = 1 - \max_K(\hat{p}_k)$$

- But – This is jumpy and noisy and this is not sensitive to tree growing

Cross Entropy

$$D = - \sum_{k=1}^K \hat{p}_{mk} \log \hat{p}_{mk}.$$

Classification Cost Function – Gini Index

- Measure of variability
- Moves smoothly
- Min Gini index desirable
- How to calculate Gini_Index
 1. Calculate Gini index for each parameter
 2. Calculate weighted sum of Gini indexes
 3. Choose the lowest Gini Index

Gini index and Deviance

- The *Gini index* is defined by

$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk}),$$

a measure of total variance across the K classes. The Gini index takes on a small value if all of the $\hat{p}_{mk}$'s are close to zero or one.

- For this reason the Gini index is referred to as a measure of node *purity* — a small value indicates that a node contains predominantly observations from a single class.

Entropy & Gini

1. entropy

$$H = -(p_1 \log_2 p_1 + \dots + p_n \log_2 p_n)$$

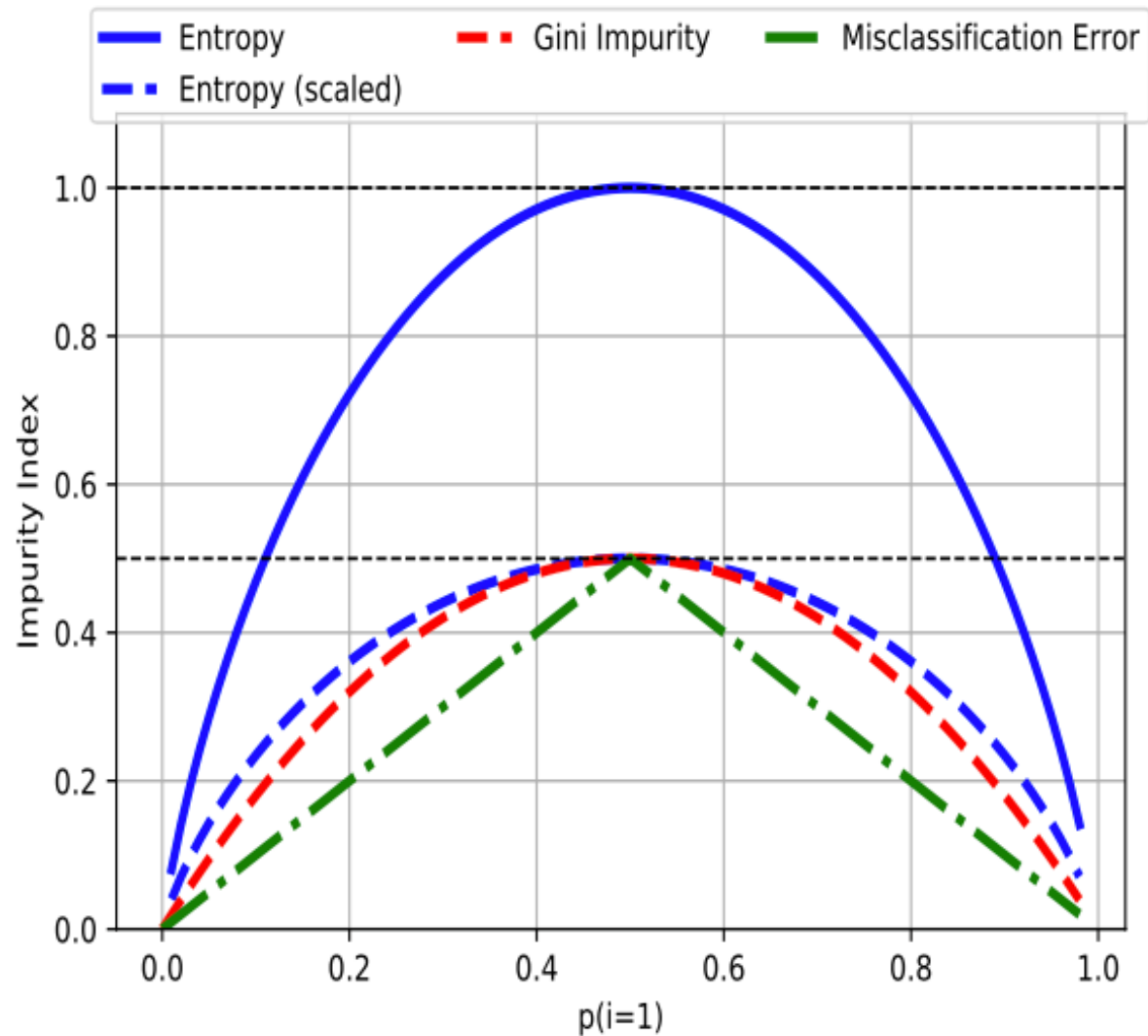
- $H = 0$: all in one class
- $H = 1$: uniform distribution

2. gini impurity

- for binary classification
- requires ranking
- a measure of mis-classification

$$\begin{aligned} \text{Gini} &= p_1(1 - p_1) + \dots + p_n(1 - p_n) \\ &= 1 - (p_1^2 + \dots + p_n^2) \end{aligned}$$

Gini vs. Entropy



Logistic regression R2

- McFadden's Pseudo R2:
 - LL(fit)
 - Calculate log odds for every point
 - Translate log odds to probability – logistic function
 - Calculate log likelihood of the data
 - LL(overall prob) replaces SS(mean) in linear reg
 - Calculate probability of the event – cancer prob/all the people
 - Or
 - Calculate odds of the event from the samples
 - Log of this is log odds based on sample
 - Then translate log odds to probability
 - Which is the same as prob of the event
 - Then calculate log likelihood based on this probability

$$R^2 = \frac{LL(overall\ prob) - LL(fit)}{LL(overall\ prob)}$$

- This R^2 is zero for bad model and it is equal to one for a perfect model

Chi squared and P_value

- Chi squared is used when we compare two categorical variables and their relationship is linear.
- P value is area under Chi squared curve with $df=1$ at point

$$2[LL(\text{fit}) - LL(\text{overall prob})]$$

Chi Square Formula



$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where

O_i = observed value (actual value)

E_i = expected value

Pros and Cons of Logistic Regression

- Logistic regression is easy to implement and interpret.
- If number of observations is less than the number of features, it will overfit
- No assumption on distribution of classes in feature space
- The boundaries constructed are linear
- Multinomial regression/multiple classes implementation and interpretation is easy
- Assuming linear dependency of dependent and independent variables
- Measures importance of each independent variable and also positive or negative correlation
- Not appropriate for non-linear problems and complex relationships
- Mostly doesn't overfit
- Extracts linear dependency of independent variables and log odd of dependent variable.

Design considerations

- Threshold doesn't need to be set at 0.5
- Don't change continuous variable to categorical, since info will be lost
- Multivariate logistic regression (Softmax) is easy to implement
- When samples are fully separated, discriminate analysis is a good option (which is simply the same approach with assumption of normal distribution for the classes), when
 - Number of samples is small
 - Classes are fully separatable

- Good reference:

<https://arunaddagatla.medium.com/maximum-likelihood-estimation-in-logistic-regression-f86ff1627b67>